

David Dongjun Yang

(424) 394-3236 | djyang0530@gmail.com | Personal Page | US Citizen | Secret Clearance

EDUCATION

Emory University

Bachelor of Science in Computer Science; GPA: 3.87/4.00

Aug 2022 – May 2026

Atlanta, GA

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, JavaScript, TypeScript, R, Bash

Focus: ML Engineer, ML Research Engineer, AI Engineer, Machine Learning, Deep Learning, Generative AI, LLMs, RAG

Methods: Computer Vision, Reinforcement Learning, Foundation Models, Self-Supervised Learning, Graph Neural Networks

Tools / Infra: PyTorch, TensorFlow, scikit-learn, LangChain, FAISS, Neo4j, SFT, LoRA, PEFT, GRPO, MLOps, AWS, Azure AI Foundry, GCP, Docker, Linux, Git, SQL

EXPERIENCE

Machine Learning Research Engineer

June 2026 – Current

Woven Solutions (DARPA & U.S. Cyber Command Programs)

McLean, VA

- Post-trained FLUX.2 [klein] **9B** and Qwen-Image-2.1 **7B** with **GRPO** against in-house VLM reward models, improving Aesthetics and Real-world Fidelity on Qwen-Image-Bench by **12%** (relative).
- Built **RL reward models** by post-training Muse Glimmer **30B** (VLM) with SFT and GRPO, and SigLIP2 1B with SFT.
- Preprocessed a **240K**-image dataset for model post-training.

AI Engineer

Jan 2026 – June 2026

Emory University Office of Admission

Atlanta, GA

- Architected a multi-layer agentic conversation system in **Azure AI Foundry** leveraging **GPT 5.4**, **Azure SQL**, **MCP**, and text-embedding-3-large; integrated 70k+ rows of admissions data.
- Deployed a Microsoft Teams bot for counselors and marketing staff, achieving **92% human-rated task success** across 700+ production queries focused on market analysis and admission strategies.

Machine Learning Engineer Intern

June 2025 – Aug 2025

Naval Information Warfare Center (U.S. DOD)

Guam, USA

- Developed an **on-device RAG** pipeline for secure, localized question answering over 1.2k classified documents by adapting LoRA fine-tuned **Gemma3 4B** and **Llama3 8B** models.
- Engineered a **hybrid retrieval pipeline** utilizing **LangChain**, combining **FAISS** chunk semantic search with **Neo4j** graph entity linking to maximize evidence coverage and grounding.
- Evaluated and deployed the system to NIWC Engineers via Ollama, achieving **85% top-k retrieval relevance** with a **7.3s end-to-end local latency** on a 200+ query benchmark.

Research Assistant

Jan 2023 – Oct 2026

UCLA & Emory University

Atlanta, GA

- **Pre-trained** an **85M**-parameter foundation model for human sensor signals utilizing **ViT** encoders and **DINO/MAE** self-supervised learning, significantly improving missing-channel inference and out-of-domain transfer.
- Designed a harmonization pipeline across 9 public **multi-modular** PSG datasets, standardizing a 12-channel montage over **166.5K** hours, **21,482 recordings**, and **20M 30-second epochs**.
- Achieved **97.3 AUROC** and **90.4 AUPRC** on sleep staging under linear probing, improving to **97.9 / 92.0** under full **fine-tuning**, alongside performance gains on apnea and hypopnea detection.
- **Post-trained** an MAE-based **10M**-parameter infant EEG **foundation model** with continuous SSL, yielding **R2=0.91** and **12.7 MSE** for precise motor-score regression.

Machine Learning Research Intern

May 2024 – Aug 2024

UNC Greensboro @ GraLNA (U.S. NSF REU)

Greensboro, NC

- Developed an fMRI **image classification Graph Neural Network** architecture for neurodegenerative disease, utilizing a novel co-embedding aggregation layer for joint node/edge connectivity.
- Extracted representations from **459 patient scans** (ADNI/PPMI databases), generating node correlation matrices and **edge functional connectivity** mappings across **116 brain regions**.
- Benchmarked the model against CNN, GCN, CRGNN, and MGNN architectures, achieving state-of-the-art results including **84.4% precision** and **76.6% F1** on ADNI under 10-fold cross-validation.

CERTIFICATIONS

Google Data Analytics Certificate	2026
Google, via Coursera	
Advanced Data Modeling	2026
Meta, via Coursera	

PUBLICATIONS & PRESENTATIONS

EEG Foundation Modeling for Early Prediction of Infant Motor Development	2026
D. Yang and R. Xiao	
<i>8th Annual Pediatric Early Career Researcher Conference, Atlanta, GA</i>	
On Pre-training and Scaling of Sleep Foundation Models	2026
Z. Shuai, Z. Xu, D. Yang, W. Wang, and Y. Yang	
<i>Proceedings of the International Conference on Machine Learning (ICML)</i>	
ConStruct: Structural Distillation of Foundation Models for Prototype-Based Weakly Supervised Histopathology Segmentation	2026
K. Le, H. Thach, A. M. Vu, T. T. K. Vo, H. H. Huynh, D. Yang, M. H. N. Le, T. H. Nguyen, et al.	
<i>Proceedings of Medical Imaging with Deep Learning (MIDL)</i>	
IMACT-CXR: An Interactive Multi-Agent Conversational Tutoring System for Chest X-Ray Interpretation	2026
T. A. Le, A. M. Vu, D. Yang, A. Awasthi, and H. Van Nguyen	
<i>Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI)</i>	
Edge-Boosted Graph Learning for Functional Brain Connectivity Analysis	2025
D. Yang, M. Abdelmegeed, J. Modl, and M. Kim	
<i>Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI)</i>	

PREPRINTS & UNDER REVIEW

Infant-Adapted EEG Foundation Modeling for Developmental Status Classification in Infants Born Preterm	2026
D. Yang, B. A. Smith, and R. Xiao	
<i>Under review — NeurIPS 2026 BrainBodyFM</i>	
DualProtoSeg: Simple and Efficient Design with Text- and Image-Guided Prototype Learning for Weakly Supervised Histopathology Image Segmentation	2025
A. M. Vu, K. P. Le, T. T. K. Vo, H. Thach, H. H. Nguyen, D. Yang, H. H. Huynh, et al.	
<i>arXiv preprint — arXiv:2512.10314</i>	
Beyond the First Read: AI-Assisted Perceptual Error Detection in Chest Radiography Accounting for Interobserver Variability	2025
A. Vutukuri, A. Awasthi, D. Yang, C. C. Wu, and H. Van Nguyen	
<i>arXiv preprint — arXiv:2506.13049</i>	

AWARDS

Clinton Joiner Career Development Award—Best Research Rapid-Fire and Poster Presentation	2026
8th Annual Pediatric Early Career Researcher Conference	
<i>Awarded for “EEG Foundation Modeling for Early Prediction of Infant Motor Development,” Arthur M. Blank Hospital, Children’s Healthcare of Atlanta, Atlanta, GA.</i>	
NSF/CRA REU Travel Grant	2025
Computing Research Association (CRA)	